DHOLE PATIL EDUCATION SOCIETY'S
**DHOLE PATIL COLLEGE OF ENGINEERING**

## Department of First year Engineering

**Course Code: ESC-102-ELE** **Subject: Basic Electrical Engineering** **Academic Year – 2025-26**

**Question Bank** **Semester -I**

| Unit –III AC FUNDAMENTAL | |
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| **S.NO:** | **QUESTIONS** |
| Que.1 | Obtain the expression for current and power, when voltage $v$ = Vm sinωt is applied across purely resistive circuit. Also draw the waveform for voltage, current & power on common X-axis. |
| Que.2 | Derive the expression for RMS value in terms of its maximum value for sinusoidally varying alternating current. |
| Que.3 | An alternating current is given by i=14.14 sin (377 t).<br>Find (a) rms value of the current (b) frequency<br>(c) instantaneous value of the current at t=3 ms<br>(d) time taken by the current to reach 10A for first time after passing through zero. |
| Que.4 | Obtain the expression for instantaneous power in case of purely capacitive circuit and by using this expression, prove that this circuit never consumes power. |
| Que.5 | Define RMS value. A sinusoidal voltage with frequency of 50 Hz has RMS value of 200 V. Write the expression for its instantaneous value and find its instantaneous value at t = 1.78 mSec and time required to reach 100 V for first time from t = 0. |
| Que.6 | Define average value and derive the expression for average value of sinusoidal current in terms of its maximum value. |
| Que.7 | The inductor has an inductance of 1 H and is connected to a 120 V 60 Hz line. How much current will flow in this circuit? |
| Que.8 | Determine the following parameters of a voltage v = 200 sin314t.<br>(a) Frequency (b) Form factor (c) Crest factor |
| Que.9 | A sinusoidal voltage V=50 Sin ωt is applied to a series RL circuit. The current in the circuit is given by I=25 Sin (ωt-530).<br>Determine (a) Apparent power (b) Power factor (c) Average power |
| Que.10 | Define the following terms.<br>(a) Waveform<br>(b) Cycle<br>(c) Time Period<br>(d) Frequency<br>(e) Instantaneous Value<br>(f) Peak Value/Maximum Value/Amplitude: |
| Que.11 | Derive the expression for average value of a sinusoidal alternating current in terms of its peak value. Also write the formula for i) Form Factor and ii) Amplitude Factor |
| Que.12 | Obtain the expression for current, when voltage v = Vm, sin ωt is applied across purely inductive circuit. |
| Que.13 | Define Form factor and Peak Factor. |
| Que.14 | Explain the concept of<br>a) in-phase b) lagging phase c) and leading phase quantities<br>with the help of suitable wave forms |